Regional Scale Mineral Exploration Through Joint Inversion And Geology Differentiation Based On Multi-physics Geoscientific Data

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SUMMARY

Modern mineral exploration focuses in underexplored regions where the terrain and geological environments make the discovery of mineral ore deposits increasingly difficult. The availability of multi-physics data is potentially useful because the complementary information contained in each data set can be integrated into a common Earth model consistent with all available geoscientific information. The goal of this study is to develop a workflow for quantitative integration of geophysical and geological data for mineral exploration. We focus on two important components of the workflow: joint inversion and geology differentiation. Joint inversion allows physical property models to constrain each other at the inversion stage, resulting in structurally similar models and enhanced correlations between inverted physical property values. Geology differentiation makes use of the jointly inverted physical property values and builds a 3D quasi-geology model that shows the spatial distribution of various geological units. The proposed workflow is first tested on a synthetic data set before being applied to a set of airborne gravity and magnetic data in central British Columbia. We have successfully identified multiple geological units that are consistent with the airborne geophysical data and prior geological information. This work provides guidance for follow-up detailed geophysical surveys in the study area and highlights the benefits of integrative interpretation of multi-physics geoscientific data.

INTRODUCTION

Geophysical methods have long been used to explore the subsurface in various geological settings and with different purposes. Advances in surveying techniques have allowed for the acquisition of larger data sets in remote regions, as well as for the simultaneous acquisition of multiple data types (Oldenburg and Pratt, 2007; Farquharson and Lelièvre, 2017). Gravity and magnetic data are especially useful for mineral exploration, but the interpretation of the data has traditionally been accomplished by performing separate inversions without taking advantage of the complementary information in these collocated data sets. This poses a particular problem, because the inverted models may be inconsistent with each other, thereby making the efforts for an unbiased and objective interpretation more challenging.

Joint inversion provides a mathematical framework that allows for simultaneous inversion of multiple types of geophysical data while at the same time permitting exchange of structural and/or petrophysical information between different physical property models (e.g., Afnimar and Nakagawa, 2002; Gallardo and Meju, 2003a; Linde et al., 2006; Colombo, 2007; Jegen et al., 2009; Moorkamp et al., 2011; De Stefano et al., 2011; Moorkamp et al., 2011; Zhdanov et al., 2012; Lelièvre et al., 2012; Haber and Holtzman Gazit, 2013; Li and Qian, 2016; Sun and Li, 2017; Zheglova et al., 2018; Colombo and Rovetta, 2018; Astic and Oldenburg, 2019). The resulting models are both structurally and petrophysically more consistent with each other, and consequently, are more suited for integrative interpretation.

We implemented joint inversions of regional scale airborne gravity and magnetic data by enforcing structural similarity between the inverted physical property models. The data were collected over central British Columbia as part of the QUEST project. We show that joint inversion can improve the structural similarity and physical property correlations of the inverted models. We also performed geology differentiation, i.e., the process of identifying and delineating various geological units (Kowalczyk et al., 2010a; Martinez and Li, 2015; Sun and Li, 2015; Devriese et al., 2017; Fournier et al., 2017; Kang et al., 2017; Melo et al., 2017), based on which we constructed a 3D quasi-geology model (Li et al., 2019a) that shows the spatial distribution of various geological units in our study area. Our work shows that joint inversions facilitates geology differentiation and helps identify geological units.

METHODOLOGY

We first perform joint inversions on the gravity and magnetic data to recover 3D distributions of density contrast and magnetic susceptibility. The objective function to be solved in the joint inversion is given by

$$\Phi(\vec{m}_1,\vec{m}_2) = \Phi_{d1} + \beta_1 \Phi_{m1} + \Phi_{d2} + \beta_2 \Phi_{m2} + \lambda \sum_{i=1}^{M} \left\| \vec{\nabla} m_1^{(i)} \times \vec{\nabla} m_2^{(i)} \right\|_2^2 \quad (1)$$

where the second subscript refers to the model (i.e. model 1 or model 2), Φ_d is the data misfit, Φ_m is the regularization (i.e. Tikhonov regularization), β is the regularization parameter, λ is the weight for the coupling term, and the last term is the cross-gradients as defined by Gallardo and Meju (2003b).

The inverted density contrast and magnetic susceptibility values are then visualized in a scatterplot and classified into different classes. The classification is based on both the features in the inverted physical property values (e.g., natural groupings, linear trends) and the existing geological information. Each class is characterized by a distinct range of density contrast and susceptibility values, and corresponds to a unique geological unit. The identified geological units are then mapped onto the 3D model space to create a representation of geology, which we refer to as a 3D *quasi-geology model*. We refer to the process of arriving at a quasi-geology model as *geology differentiation*.

SYNTHETIC TEST

A synthetic example is first presented to illustrate the workflow for geology differentiation based on joint inversions. The example consists of gravity and purely induced total magnetic intensity (TMI) data simulated from four causative bodies that are cube-shaped. The simulated data is shown in figure 1. For brevity, the causative bodies and inverted models are not shown. The scatterplot of the jointly recovered density contrast and magnetic susceptibility values are shown in Figure 2. We have included the scatterplot from separate inversions as a comparison with joint inversions. We note that the scatterplot from separate inversions is excessively smooth and scattered, making it less ideal to perform geology differentiation. In contrast, the scatterplot from joint inversions shows clear trends that are indicative of the different causative bodies.

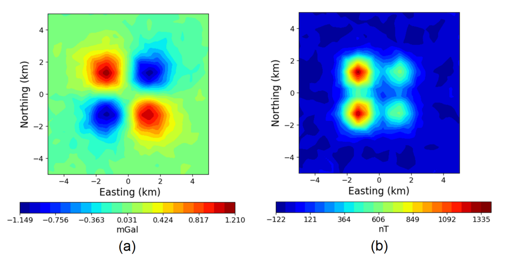

Figure 1: Simulated (a) gravity data, and (b) TMI data.

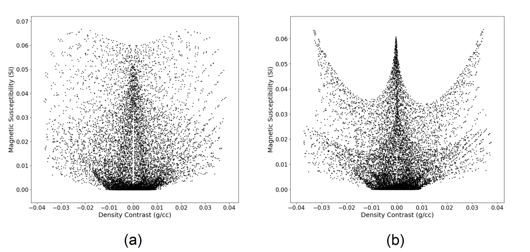

Figure 2: Scatterplots of the recovered density contrast and magnetic susceptibilities from (a) separate and (b) joint inversions.

Geology differentiation

Interpretation of the recovered physical property models from 3D inversions has traditionally relied on the identification of specific features (such as local highs and lows) in the models themselves. However, as pointed out by Li et al. (2019b), to maximize the value of geophysical data, we need an additional step to move the inverted physical properties to approximations of geology. In this section, we performed geology differentiation on the jointly inverted models. The classification is based on the fact that the four causative bodies consist of distinct density contrast and magnetic susceptibility values that range from "low" to "high" as shown in figure 3. In Figure 2, we observe a cone-shaped feature in the central top area of the crossplot as well as four wing-shaped features on both sides. Consequently, we classified the inverted values into five units, as shown in Figure 3. The classification results in the crossplot were then mapped into 3D spatial domain. Figure 4 shows the 3D quasi-geology model where the four identified units corresponding to the causative bodies are recovered at roughly the correct locations and depths. However, we observe traces of units C and D on top of units A and B, which occurs as a result of the smoothness-based regularization used for the inversions.

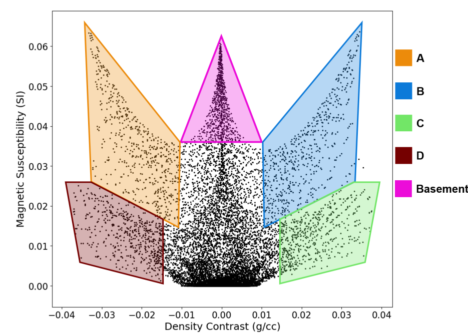

Figure 3: Classification of the jointly inverted values.

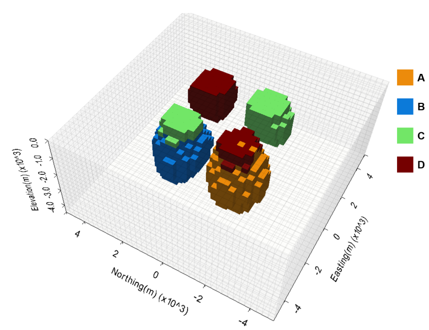

Figure 4: Three-dimensional view of the causative bodies identified through geology differentiation.

APPLICATION TO THE QUEST FIELD DATA

We applied the same workflow consisting of joint inversions followed by geology differentiation to the airborne gravity and magnetic data collected over a central region in British Columbia as part of the QUEST project (Geoscience BC, 2007). The motivation for our work is to improve on the work in Kowalczyk et al. (2010b) and Phillips et al. (2009). They have inverted the airborne gravity, magnetic, and electromagnetic (EM) data of the QUEST project and subsequently interpreted the inverted results through a classification process. However, the geologic reasoning behind how each class was defined remains ambiguous or unstated. Additionally, the classification was based on the results from separate inversions, which poses a limitation as discussed above.

Geological setting

The study area focuses on the Quesnel terrane located in British

Columbia, Canada. The Quesnel is an early Mesozoic volcanic arc and contains significant Cu-Mo and Cu-Au porphyry deposits (Logan and Mihalynuk, 2014). However, most of the central portion of the Quesnel terrane is covered by a ubiquitous Quaternary glacial sedimentary cover that has impeded exploration efforts.

Geophysical inversions

In this study, we use the gravity and magnetic data after regional field removal, as processed and made available in the Geoscience BC Report 2009-15 (Phillips et al., 2009). We apply the same methodology for joint inversions used in the synthetic example. Specifications of the inversion parameters and other details have been omitted for brevity. We also note that we allowed for negative susceptibilities in the inversions, as physical rock properties reported in the Geoscience BC Report 2013-14 by Mitchinson et al. (2013) indicate significant remanence in some of the rock samples. For simplicity, we make the assumption that all remanent magnetizations are collinear (i.e., either parallel or anti-parallel) to the present-day geomagnetic field, and thus, negative susceptibilites are attributed to polarity reversals. We point out that we had no trouble fitting the observed magnetic data, which is a strong indication that introducing more possibilities of magnetization directions is not required by the magnetic measurements.

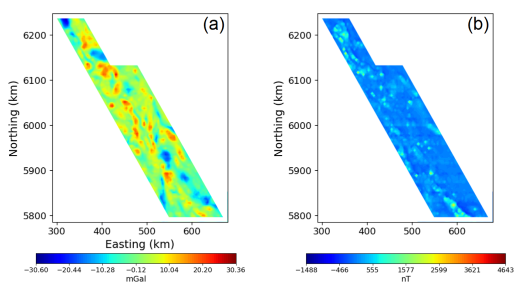

Figure 5: (a) Gravity data after regional field removal. (b) TMI data after regional field removal.

Geology differentiation

Before we classify the inverted physical property values, we need to determine the expected ranges for different geological units in the region. We summarized the inverted values in a crossplot and identified 10 large groups as shown in figure 6. The expected physical property value ranges for the 10 groups are determined by correlating the inverted physical property anomalies with the geological unit described in Cui et al. (2017) in the "BC Digital Geology" repository as well as the physical rock properties in Mitchinson et al. (2013). We refer to the information in Figure 6 as *conceptual scatterplot*, which contrasts with the *actual scatterplot* shown in Figure 7 because the former represents our expectations based on previous work before inversions were implemented and the latter summarizes what we actually obtained after inversions. The conceptual scatterplot is important as it serves as a guide for geology differentiation. We point out that the final classification in an actual scatterplot are very likely to be different from the classification of physical property values in a conceptual scatterplot. This is understandable because the former is based on recovered physical property values from smoothness regularized inversion, whereas the latter is mostly based on surface geology and limited physical property measurements.

The conceptual scatterplot crystallizes our understanding of the physical properties of various geological units. Once the conceptual scatterplot was developed, we applied the understanding to the scatterplot of jointly inverted values as shown in figure 7. We identified 17 units based on the natural grouping and linear features among the jointly inverted values as well as the conceptualization from the conceptual scatterplot. In Figure 8, we show the jointly inverted density (top row), susceptibility (middle row) and quasi-geology models (bottom row) from two different depths.

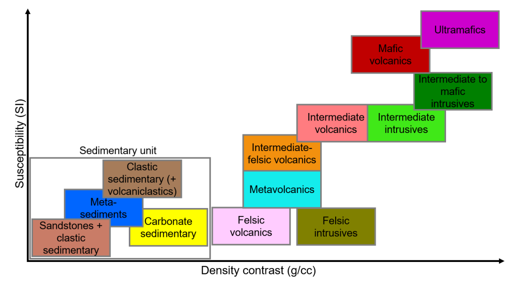

Figure 6: Conceptual scatterplot of expected density contrast and magnetic susceptibility values.

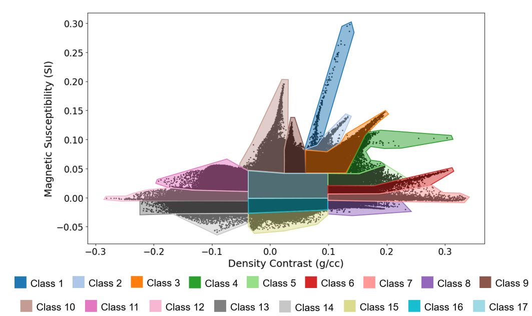

Figure 7: Classification of jointly inverted values, where each class is indicated by a polygon of unique color.

For purposes of brevity, we discuss only classes that are correlated with known geology or mineral deposits. Most notable are Classes 6 and 7 (high densities, low susceptibilities), which correlate spatially with intrusions associated with the Mount Milligan deposit, Hogem batholith, and the Trembleur ultramafite unit. The low susceptibility in the Mount Milligan deposit was explained by Oldenburg et al. (1997), where there is an anticorrelation between susceptibility and mineralization. High gold values in the Mount Milligan deposit correspond to low susceptibilities, likely due to a second stage hydrothermal alteration event that consumed magnetite. The same is expected for other areas in the region as identified in Classes 6 and 7.

Class 12, with very low densities and susceptibilities, is also noteworthy as it correlates spatially with granitic and other felsic intrusions as shown in Figure 9, such as the Bayonne

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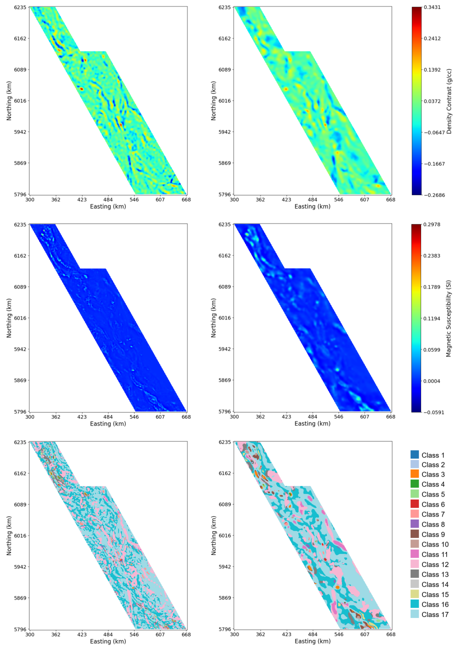

Figure 8: The columns represent horizontal slices of the models at depths (left) z = 250 m, (right) z = 4750 m. The rows represent (top) density contrast, (middle) magnetic susceptibility, (bottom) quasi-geology models.

plutonic suite, the Germansen batholith, and portions of the Hogem plutonic suite. Also worth noting is that class 11, which has the same densities as class 12 but slightly higher susceptibilities, along with class 12, contain the Granite Mountain batholith, which is host to the Gibraltar deposit. Class 10, which has low densities but high susceptibilities, correlates spatially with portions of Mount Polley in the South.

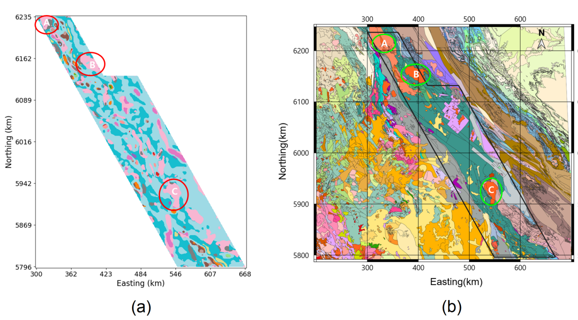

Figure 9: (a) The locations of some of the clusters corresponding to class 12 (red ellipses) on a horizontal slice at elevation z = -4750m of the quasi-geology model. (b) A geology map with the (A) Hogem plutonic suite, (B) Germansen batholith, (C) Bayonne plutonic suite encircled in green ellipses.

As a final recommendation, we identify Classes 6, 7, and 8 as areas of interest for future work. We note that these classes are correlated with known mineral deposits in several regions as shown by the red ellipses in Figure 10. What is interesting is that these classes are also located in some other areas as indicated by the blue squares in Figure 10. Geological descriptions in these areas are lacking, likely as a consequence of the Quaternary glacial sedimentary cover. However, these areas share similar physical properties with the known deposits (red ellipses) and belong to the same classes as these deposits. We, therefore, identify them as future target areas worth further exploring.

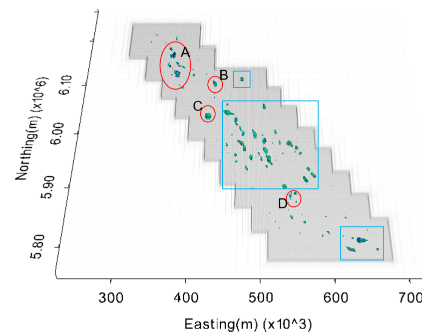

Figure 10: In red ellipses are known formations: (A) Lorraine, Takla-Rainbow, Kwanika, (B) Mount Milligan, (C) Trembleur Ulframafites, (D) Mouse Mountain. The blue squares indicate areas that were classified into the same classes and represent areas of interest for future work.

CONCLUSIONS

We have developed a workflow for constructing 3D quasi-geology models. Two important components of the workflow are joint inversion and geology differentiation. Our work shows that joint inversion leads to structurally more consistent models and facilitates geology differentiation. The application of the proposed workflow on the QUEST data has led to the identification of notable geological units associated with known mineral deposits in the region. The 3D quasi-geology model also helps identify a few underexplored regions which are covered by a thick layer of glacial sediments but share similar physical property values as the known mineral deposits. We have identified these regions as the target zones for future exploration efforts. Our work has demonstrated the benefits of integrated quantitative interpretations of geoscientific data. We are confident that the proposed workflow can be applied to many other areas for applications including but not limited to mineral exploration.

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